

MAESTRO INFRASTRUCTURE SETUP IN AWS

This document provides the information on the configuration of an instance in AWS Cloud for on premise installation of Maestro.

The described setup is necessary to enable necessary access and performance for Maestro application.

The infrastructure setup in AWS includes several steps:

- Maestro Instance Setup
- Certificate Validation
- OnPrem Infrastructure Deployment

1 MAESTRO INSTANCE SETUP

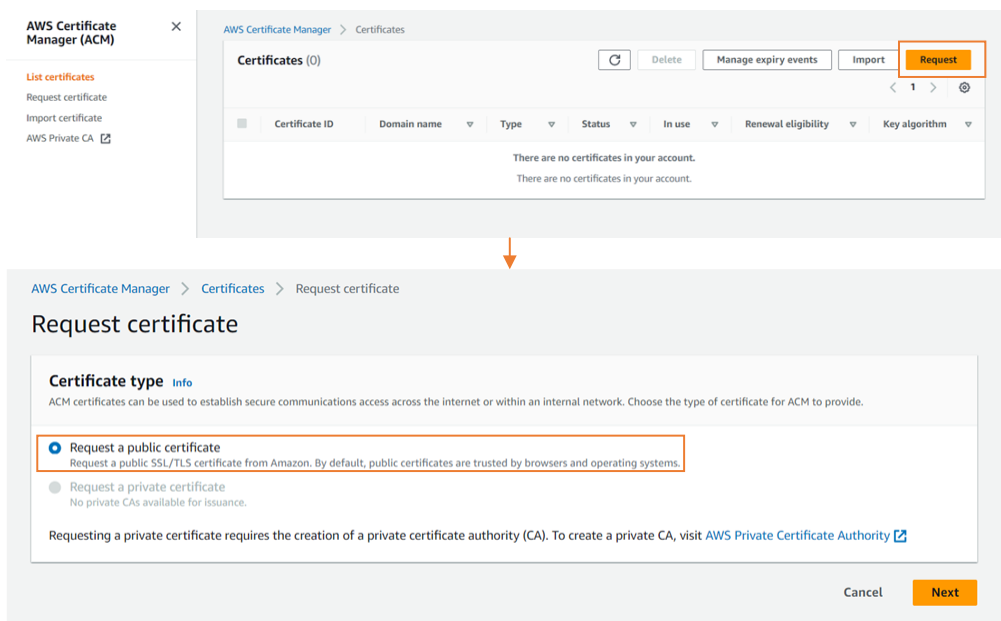
1.1 Maestro Instance Configuration

- **Minimum instance type:** m7a.large (2 vCPU, 8 GB RAM)
- **Disk size:** 100 GB
- **Operating System:** Ubuntu 22.04

1.2 Maestro Instance Setup

To setup an instance where Maestro will be installed, complete the following steps:

1. Navigate to **AWS Certificate Manager**, click **Request** and select **Request a public certificate**. This certificate needed to enable https access for Maestro.



- Specify a fully qualified domain name and request certificate as shown below:

AWS Certificate Manager > Certificates > Request certificate > Request public certificate

Request public certificate

Domain names
Provide one or more domain names for your certificate.

Fully qualified domain name [Info](#)
maestro-onprem.maestro3.community

[Add another name to this certificate](#)

You can add additional names to this certificate. For example, if you're requesting a certificate for "www.example.com", you might want to add the name "example.com" so that customers can reach your site by either name.

Validation method [Info](#)
Select a method for validating domain ownership.

☒ **DNS validation - recommended**
Choose this option if you are authorized to modify the DNS configuration for the domains in your certificate request.

☐ **Email validation**
Choose this option if you do not have permission or cannot obtain permission to modify the DNS configuration for the domains in your certificate request.

Key algorithm [Info](#)
Select an encryption algorithm. Some algorithms may not be supported by all AWS services.

☒ **RSA 2048**
RSA is the most widely used key type.

☐ **ECDSA P 256**
Equivalent in cryptographic strength to RSA 3072.

☐ **ECDSA P 384**
Equivalent in cryptographic strength to RSA 7680.

Tags [Info](#)
To help you manage your certificates, you can optionally assign your own metadata to each resource in the form of tags.

No tags associated with this resource.

[Add tag](#)

You can add 50 more tag(s).

Cancel Previous **Request**

- Once the certificate is requested it will be displayed as pending validation. Find it in the Certificates list:

AWS Certificate Manager > Certificates

Certificates (1) [Refresh](#) [Delete](#) [Manage expiry events](#) [Import](#) [Request](#)

< 1 > [Settings](#)

☐	Certificate ID	Domain name	Type	Status	In use	Renewal eligibility	Key algorithm
☐	8c24aed9-ecf2-4438-b1a4-b23c8001c0ed	maestro-onprem.maestro3.community	Amazon Issued	Pending validation	No	Ineligible	RSA 2048

- Open the created certificate entity and scroll down to the Domains section:

Domains (1) [Create records in Route 53](#) [Export to CSV](#)

< 1 >

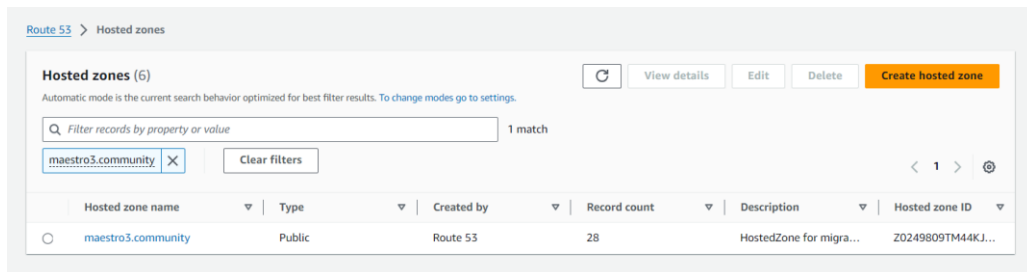
Domain	Status	Renewal status	Type	CNAME name	CNAME value
maestro-onprem.maestro3.community	Pending validation	-	CNAME	_74b1dc6427b10d9b12c054dbfef01c81.maestro-onprem.maestro3.community.	_a5fa20578a97a10c94ee34a50d7252cd.mhbtspndnt.acm-validations.aws.

- Copy "CNAME name" and "CNAME value" – it will be needed in following step to validate certificate.

2 CERTIFICATE VALIDATION

In order to validate a certificate, follow the steps below:

1. Navigate to **Route53**, select **Hosted Zones** and select a hosted zone related to fully qualified domain name specified during certificate request



2. Create a record to verify the certificate (Record name and Value can be obtained from step 5 of Maestro Instance Setup section – these are “CNAME name” and “CNAME value” respectively)

When the record is created, validation can take some time. Once the validation is completed, the certificate status will be changed to “Issued”:

	Certificate ID	Domain name	Type	Status	In use	Renewal eligibility	Key algorithm
☐	8c24aed9-ecf2-4438-b1a4-b23c8001c0ed	maestro-onprem.maestro3.community	Amazon Issued	Issued	No	Ineligible	RSA 2048

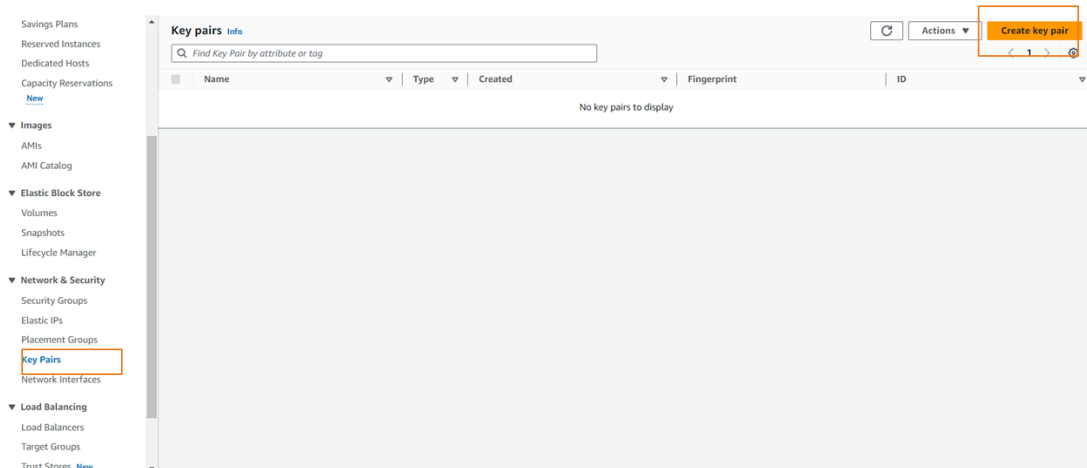
3 DEPLOY MAESTRO ONPREM INFRASTRUCTURE

When installed in AWS, Maestro also needs additional infrastructure. The deployment is performed according to the steps listed below in this section.

3.1 Create a Key Pair

To create a Key Pair to be used to connect to the instance, follow the steps below:

1. Navigate to **EC2 -> Key Pairs -> Create Key Pair**



2. Set the key name as “maestro-service-key” and change Private key file format to “.pem”. After creating key will be downloaded.

Create key pair

Key pair

A key pair, consisting of a private key and a public key, is a set of security credentials that you use to prove your identity when connecting to an instance.

Name

maestro-service-key

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA

☐ ED25519

Private key file format

☒ .pem

For use with OpenSSH

☐ .ppk

For use with PuTTY

Tags - optional

No tags associated with the resource.

Add new tag

You can add up to 50 more tags.

Cancel

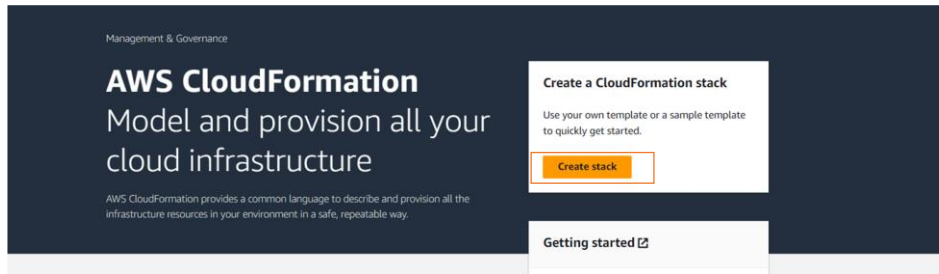
Create key pair

3.2 Launch AWS CloudFormation template

The necessary infrastructure is deployed via an AWS CloudFormation template. To do this, complete the following steps:

Please use the template provided together with this instructions.

1. Go to **CloudFormation** -> **Create Stack**



2. Choose **“Upload a template file”** as template source and upload the file provided with this instruction.

Prerequisite - Prepare template

Prepare template
Every stack is based on a template. A template is a JSON or YAML file that contains configuration information about the AWS resources you want to include in the stack.

☒ Template is ready ☐ Use a sample template ☐ Create template in Designer

Specify template
A template is a JSON or YAML file that describes your stack's resources and properties.

Template source
Selecting a template generates an Amazon S3 URL where it will be stored.

☐ Amazon S3 URL ☒ Upload a template file ☐ Sync from Git - new

Upload a template file
Choose file

maestro-onprem-aws-infra.yml
JSON or YAML formatted file

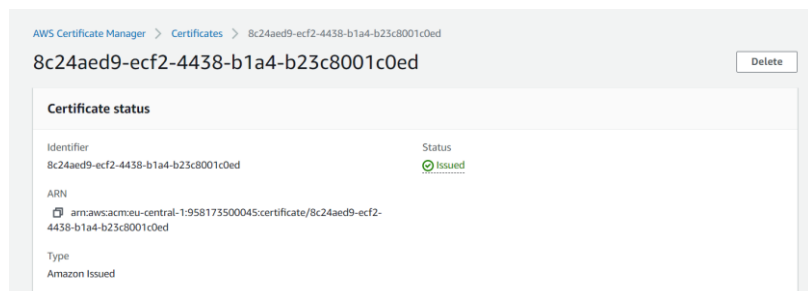
S3 URL: https://s3.eu-central-1.amazonaws.com/cf-templates-18gqapog6vkk-eu-central-1/2024-02-22T143444.215Zc0z-maestro-onprem-aws-infra.yml

View in Designer

Cancel Next

3. Specify the following parameters for the template launch:

- Stack name: maestro
- Certificate: ACM certificate ARN created in previous step. To obtain it, navigate to AWS Certificate Manager, open created certificate and copy ARN



- **KeyName:** The key created on step 3.1

Stack name

Stack name can include letters (A-Z and a-z), numbers (0-9), and dashes (-).

Parameters

Parameters are defined in your template and allow you to input custom values when you create or update a stack.

Certificate
Certificate arn to use

EnvPrefix
Environment prefix

EnvSuffix
Environment suffix

EnvironmentName
An environment name that will be used for tags

KeyName
Key Pair for EC2

4. Leave other parameters default, scroll down, and click Next. At “Configure stack options” section keep everything default and move to the **Next** section.

CloudFormation > Stacks > Create stack

Step 1
[Create stack](#)

Step 2
[Specify stack details](#)

Step 3
Configure stack options

Step 4
[Review and create](#)

Configure stack options

Tags
You can specify tags (key-value pairs) to apply to resources in your stack. You can add up to 50 unique tags for each stack.

No tags associated with the stack.

[Add new tag](#)
You can add 50 more tag(s)

Permissions

IAM role - optional
Choose the IAM role for CloudFormation to use for all operations performed on the stack.

IAM role name [Remove](#) [Refresh](#)

Advanced options
You can set additional options for your stack, like notification options and a stack policy. [Learn more](#)

► **Stack policy**
Defines the resources that you want to protect from unintentional updates during a stack update.

► **Rollback configuration**
Specify alarms for CloudFormation to monitor when creating and updating the stack. If the operation breaches an alarm threshold, CloudFormation rolls it back.

► **Notification options**

► **Stack creation options**

Cancel [Previous](#) [Next](#)

5. Scroll down and press Submit

The screenshot shows the 'Review and create' page for a new AWS CloudFormation stack. The page is divided into two main sections: 'Step 1: Specify template' and 'Step 2: Specify stack details'.

Step 1: Specify template

- Prerequisite - Prepare template:** This section shows the template URL and the stack description. The template URL is `https://s3.eu-central-1.amazonaws.com/cf-templates-18gapog6vsk-eu-central-1/2024-02-22T162944.671Zzwc-maestro-onprem-aws-infra.yml`. The stack description is 'AWS CloudFormation template for VPC'.
- Template:** This section shows the template URL and the stack description.

Step 2: Specify stack details

- Notification options:** This section shows the SNS topic ARN. There are no notification options defined.
- Stack creation options:** This section shows the timeout and termination protection. The timeout is set to '-' and the termination protection is 'Deactivated'.

At the bottom of the page, there are buttons for 'Cancel', 'Previous', and 'Submit'.

Infrastructure creation may take some time. After all resources are created, the stack will be in `CREATE_COMPLETE` state.

The screenshot shows the AWS CloudFormation console. On the left, there is a list of stacks. The stack named 'maestro' is highlighted with a red box. Its status is 'CREATE_COMPLETE'.

On the right, the 'maestro' stack is selected, and the 'Overview' tab is active. The overview shows the following details:

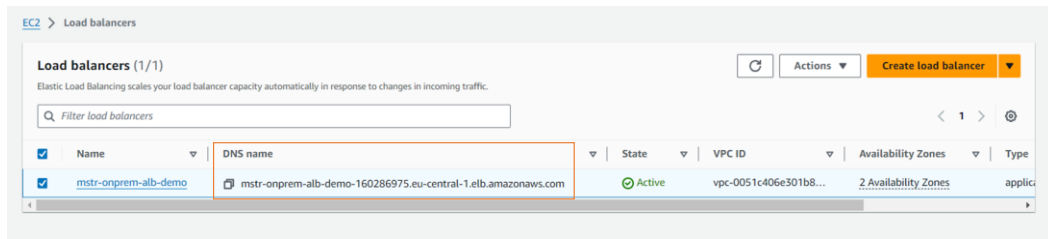
- Stack ID:** `arn:aws:cloudformation:eu-central-1:218678211584:stack/maestro/ed9ce160-d1a3-11ee-9e41-06af7ac1d98d`
- Description:** AWS CloudFormation template for VPC
- Status:** `CREATE_COMPLETE`
- Status reason:** -
- Root stack:** -
- Parent stack:** -
- Created time:** 2024-02-22 19:00:49 UTC+0200
- Deleted time:** -
- Updated time:** -

3.3 Create Route53 record for Load Balancer

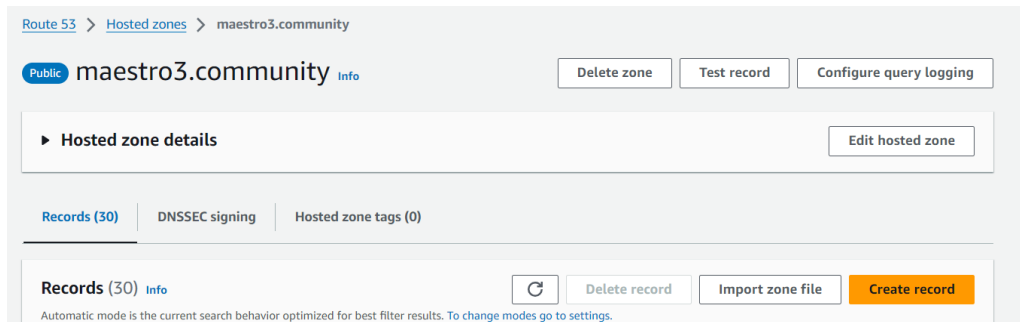
To create a Route53 record for Load Balancer, do the following:

1. Navigate to **EC2 -> Load Balancer**. Retrieve Load Balancer DNS name. it will be used in following step.

Maestro – Instance Setup in AWS



2. Navigate to **Route53** -> **Hosted Zones**. Select Hosted zone used for certificate creation and press Create record:



3. Use your certificate from earlier steps as Record name. Choose “CNAME” for Record type. In value field paste Load Balancer DNS retrieved from previous step and click Create record.

